

MID Semester of 3rd Sem B.Tech.

MID-SEMESTER EXAMINATION, SEPTEMBER 2025

COURSE CODE: EPMT301

COURSE TITLE: MATHEMATICS FOR MACHINE LEARNING

TIME: 1.30 HRS

MAX. MARK: 20

Note:- Attempt all questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. NO	QUESTION	MARKS	CO
1a	$T: R^3 \rightarrow R^3$ defined by $T(x, y, z) = (2y + z, x - 4y, 3x)$. Determine the matrix of the linear transformation T with respect to the standard ordered basis of R^3 .	2	CO1
1b	Determine whether the given set $W = \{(x, y, z) \in R^3 : x = z + 2\}$ is a subspace of R^3 . Justify your answer.	2	CO1
2a	Determine the linear mapping $T: R^3 \rightarrow R^3$ that maps the basis vectors $(2, 1, 1), (1, 2, 1), (1, 1, 2)$ of R^3 to the vectors $(1, 1, 1), (1, 1, 1), (1, 1, 1)$ respectively.	2	CO1
2b	Determine whether the set $S = \{(1, 3, 2), (1, -7, -8), (2, 1, -1)\}$ is linearly dependent or independent. Justify your answer.	2	CO1
3a	The linear transformation $T: P_2(R) \rightarrow M_{2 \times 2}$ is defined by $T(f(x)) = \begin{bmatrix} f(1) - f(2) & 0 \\ 0 & f(0) \end{bmatrix}$. Find the basis and dimension of range T .	2	CO1
3b	Let $u = (1, 3), v = (2, 1) \in R^2$. Find $w \in R^2$ such that $\langle w, u \rangle = 3, \langle w, v \rangle = -1$, where $\langle, \rangle$ is the standard inner product on R^2 .	2	CO2
4a	Discuss the nature of the following quadratic form $Q(x, y, z) = x^2 + 2y^2$.	2	CO2
4b	Product of two eigen values of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ is 16. Find the third eigen value.	2	CO2
5a	Use Gram-Schmidt process, convert the basis $\{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ of the Euclidean space R^3 with the standard inner product into an orthogonal basis.	2	CO2
5b	Show that $S = \{(2, -1, 3), (-1, 1, 1), (-4, -5, 1)\}$ is orthogonal basis of R^3 . Find the orthonormal basis of R^3 .	2	CO2